# Internship Final Presentation

Yen Nguyen

# INTRODUCTION

Hello!
I’m Yen!

# FLOWMETRIC

Flowmetric is a pioneering company dedicated to enhancing environmental protection and economic resilience through advanced corporate water management solutions.

Mission: “Empowering water-intensive organizations with advanced technology to navigate the challenges of climate change and ensure a sustainable water future”.

Goal: To collaborate with water-intensive companies, enhancing their water efficiency and ensuring compliance with stringent environmental regulations and international standards.

# Our Team

## Wilder
CEO

## Nam
COO

## Yen
Intern

# Responsibilities

## 01 Market Research

-Extensive market research on market landscape and the position of the company within it.
- Looking at other comparable companies, which provided valuable insights into the climate risk management space.

bluesign®

waterplan

Ceres

### Southeast Asia Market

#### Textiles

Significant water challenges, including high water consumption, pollution, and regulatory pressures.
Demin producers and expand to other textile manufacturers.

#### Food & Beverage

Beverage production, brewing and bottling.
Food processors: maintaining high water quality and availability is critical for food processors.

#### Industrials

Rely heavily on water for cooling and other operational needs.
Cooling system
Water recycling

#### Agriculture

Agriculture in Southeast Asia consumes a substantial amount of water.
Irrigation
Manage water resources

| Company | Compliance Mandate | Contact Name | Contact position | Link | rejected/rel | Note |
|---|---|---|---|---|---|---|
| ıınsung ectronics etnam | | | | SamSungVietnam | | Region: Bac Ninh, Thai nguyen |
| ıxconn ın Hai ecision ert Products etnam Co., d. | m/reports/CSR/EN/2022/Hontai_2022_Sustainability_Report_EN- | | | https://www.foxconn.com.vn/contact AnCham Vietnam-Intel Products Vietnam | | fabrication, PCB fab, semiconductor assembly,cooling Wafer fabrication Cleaning and cooling(cleaning |
| ettel High chnology lustries rporation | https://viettel.com.vn/en/news-events/news/viettel-honored-as-a-sustainable-brand- | | | https://viettelhightech.vn/en/contact#map | | high-tech electronics manufacturing |
| i Electronics etnam | https://www.lg.com/global/sustainability/circularity/ | | | | | electronic components, display panels and semiconductor |
| sign etnam | https://www.renesas.com/us/en/company/sustainability | | | https://www.renesas.com/us/en/about | | Subsidiary of Renesas Electronics Corporation |
| miconducto ietnam | Compliance: REACH, green partner | | | https://www.onsemi.com/company/about-onsemi/locations/dong-nai-province-vietnam | | Dong Nai and Binh Duong Province |

# Responsibilities

## Grant Research

02

-Focusing on identifying grants that support water sustainability and innovative companies working towards green initiatives.
-Looking at different grant opportunities and understanding their alignment with the company goal.

fundsforNGOs
Grants and Resources for Sustainability

United Nations Development Programme

P4G
Pioneering Green Partnerships, Investing in Impact

USAID
FROM THE AMERICAN PEOPLE

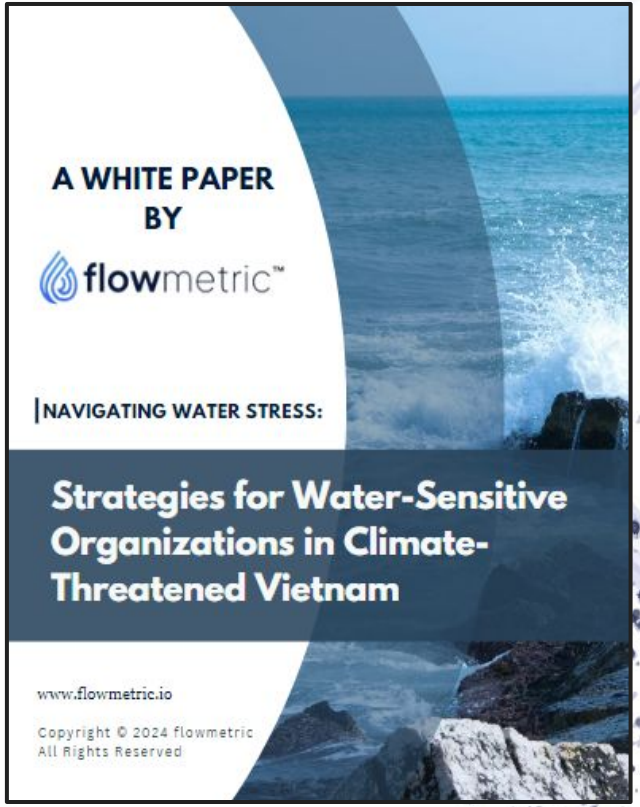
| Grant Name | Administrator | Note | Criteria | Amount |
|---|---|---|---|---|
| Special Climate Change Fund (SCCF) | World Bank | | Work with GEF (global environmental facility) Partner Agency to submit project proposals. Projects must be driven by the country, consistent with national | contact gra (request) |
| Green Climate Fund | World Bank | To receive money directly from the Green Climate Fund, entities need to be accredited as an "Accredited Entity" or AE. AEs can | Accredited Entities are responsible for monitoring and evaluating the projects they implement, as well as reporting their activities to GCF. | contact gra (request) |
| Green Climate Fund - Project Preparation Facility | World Bank | | Funding proposals developed with the support of the Project Preparation Facility should be submitted to the GCF Board within two years of the approval of a PPF request. | to USD 1.5 m |
| Water Resilience Trust Fund under | | ADB website for the | Support: Project Project preparation | contact gra (request) |

# Responsibilities

03

## White Paper

-Working on a detailed report that explored the challenges and opportunities faced by water-sensitive organizations operating in Vietnam.
-Assist with the formatting of the document.



# Responsibilities

## 04 Financial Impact

-Understanding of translating physical water risks and operational risks into measurable value.
-Involve analyzing how water scarcity and other water-related issues affect the financial performance.

| Parameters |
|---|
| *Revenue: |
| *Fixed Assets |
| *Opex: |
| *Cost of Capital |
| Cost of water per M/3: |
| Total Annual Water Bill: |
| Total Annual Wastewater Bill: |
| Total Water Used (M3) |
| Total Wastwater Discharged |
| *Shadow Price of Water M/3 |

**Scarcity Risk**

| Scenario | Annual Revenue ($) | # of D Outage |
|---|---|---|
| Low Scenario | 300,000,000.00 | |
| Medium Scenario | 300,000,000.00 | |
| High Scenario | 300,000,000.00 | |

water-related damage

**Flood Risk**

| Scenario | % of Asset Non-Performance | Fixed A |
|---|---|---|
| Low-Low | 1% | 650 |
| Medium-Low | 2% | 650 |
| High-Low | 3% | 650 |

**Water inefficiency**

| Scenario | Water Tariff ($) | M/3 lost |
|---|---|---|
| Low loss scenario | 0.51 | |
| High loss scenario | 0.51 | |

saving

**Compliance Risk** cost of compliance

| Substance | Current Wastewater Treatment Cost ($) | Addition Treatme Infrastru |
|---|---|---|
| Lead | 38,891.00 | |
| Nickel | 38,891.00 | |
| Total Chromium | 38,891.00 | |

# Other Tasks

## German Manufacture Research

| Company Name | Industry Name | Location |
|---|---|---|
| Infineon Technologies AG | Semiconductor Manufacturing | Am Campeon 1-<br>85579 Neubiberg<br>Germany |
| Elmos Semiconductor | Semiconductor Manufacturing | Heinrich-Hertz-S<br>44227 Dortmund<br>Germany |
| Otto Graf | Plastic Manufacturing | Carl-Zeiss-Straße<br>79331 Teningen,<br>Germany |
| Festo SE & Co. KG | Automation Machinery Manufacturing | Ruiter Str. 82, 73<br>Esslingen am Ne<br>Germany |
| HYDAC Group | Machinery Manufacturing | Industriestraße, 6<br>Sulzbach/Saar, G |
| ProMinent Group | Industrial Machinery Manufacturing | Im Schuhmacher<br>5-11, 69123 Heid<br>Germany |
| Bellmer | Machinery Manufacturing | Hauptstraße 37-4<br>75223<br>Niefern-Öschelbu<br>Germany |
| KSB | Industrial Machinery Manufacturing | Johann-Klein-Str<br>67227 Frankenth<br>Germany |
| Pepperl+FUCHS | Automation Machinery Manufacturing | Lilienthalstraße 1<br>68307 Mannhein<br>Germany |
| Zchimmer & Schwars | Chemical Manufacturing | Chemnitztalstraß<br>09217 Burgstädt,<br>Germany |
| FRIWO | Appliances, Electronics Manufacturing | Von-Liebig-Straf<br>48346 Ostbevern<br>Germany |

## ASEAN state of water disclosure

- I had the opportunity to connect and network with organizations: ADB Bank, USAID
- Reaching out to these organizations and exploring potential collaborations and opportunities.

# Contributions

## Market and Grant Research

Provided actionable insights that helped shape strategic planning and market positioning.

| Company | Compliance Mandate | Contact Name | Contact position | Link | Projected sale | Note | About | Date |
|---|---|---|---|---|---|---|---|---|
| Vinamilk | | | | https://www.vinamilk.com.vn/vi/don-vi-thanh- | | Food & Beverage | Dairy production, milk | |
| Vinh Hao | | | | https://www.vinhhao.com.vn/lien-he | | Food & Beverage | Mineral Water, bottling mineral water | |
| TH Group | | | | https://thmilkfood1.valhalla10.stage.job | | Food & Beverage | Dairy production, milk | |
| ACECOOK | | | | https://acecookvietnam.vn/chi-nhanh/ | | Food & Beverage | Producer of instant noodles | |
| PepsiCo Vietnam | | | | https://www.suntorypepsico.vn/ | | Food & Beverage | Beverage Drink | |
| Nestle Vietnam | | | | https://www.nestle.com.vn/vi/aboutus/nestlevietnam | | Food & Beverage | Coffee Market, MILO, MAGGI | |
| Tan Hiep Phat | | | | https://www.thp.com.vn/ve-chung-toi/ | | Food & Beverage | Soft Drinks, energy drinks, and bottled water | |

| Priority | Company | Compliance Mandate | Contact Name | Contact position | Link | Projected sale | Note | About |
|---|---|---|---|---|---|---|---|---|
| | Delight Vietnam Industrial Company Limited | Bluesign | | | https://www.delightindustrial.com/en | | | Initial focus on producing greige textiles. Greige refers to unfinished textiles. |
| | MIRAE Fiber Joint Stock Company | Bluesign | | | http://miraejsc.com/en/ | | | Padding, quilting, beddings, mattresses, and garment machinery |
| | ERE Saigon CO., LTD | Bluesign | | | http://erewebbing.com/ | | | Produce webbing, elastic band, string |
| | Continental Textiles Co., LTD | Bluesign | | | https://continentaltextile.com.vn/contact | | | Hiigh-quality textile, fashion apparel to durable materials |

Provided a comprehensive resource for the team to pursue funding opportunities.

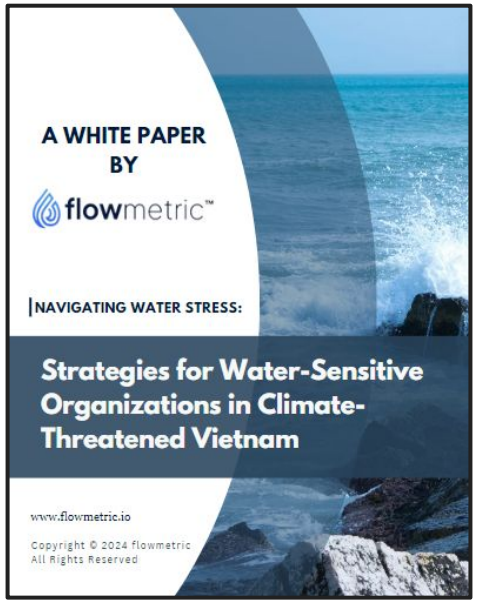
| Grant Name | Administrator | Note | Criteria | Amount | Deadline | Date Sent | Status | Grant Website | Description |
|---|---|---|---|---|---|---|---|---|---|
| FY2021 to FY2024 NOAA Broad Agency Announcement (BAA) | Department of Commerce | | Request applications for special projects and programs associated with NOAA's strategic plan and mission goals, as well as to provide the general public with information and guidelines on how NOAA | contact grant (request) | 30-Sep-24 | | | HIGHERGOV: https://www.highergov.com/grant-opportunity/fy2021to-fy2024-noaa-broad-agency-announcement-baa-32926/#overview | This Broad Agency Announcement is a m encourage research, education and outrea projects, or sponsorships that are not addr NOAA's competitive discretionary pr |
| USAID Development Innovation Ventures (DIV) Program | Agency for International development | | To be competitive under this solicitation, applications must be fully responsive to all directions within this APS document. Note that USAID may not award to an applicant unless the | $25,000 - $15,000,000 | 31-Oct-24 | | | USAID Development Innovation Ventures (DIV) Program | USAID's goal of dramatically improving lives of the poor in developing countries, a ability to adhere to DIV's core princip provideexcellent value to the U.S. gove substantially contribute to these obj |
| P4G Call For Partnership Applications | Pioneering Green Partnerships, Investing in impact P4G | | Applicants must provide services or products that contribute to poverty alleviation, gender equity and economic growth in one | USD 100,000 - 500,000 between 18-24 months | 9/8/2024 | | | P4G Call for Partnerships | P4G competitively selects partnerships early-stage climate entrepreneurs who in their seed or post-seed early growth sta to series A investor readiness |

| Grant Name | Administrator | Note | Criteria | Amount | Deadline | Date Sent | Status | Grant Website | Description |
|---|---|---|---|---|---|---|---|---|---|
| Special Climate Change Fund (SCCF) | World Bank | | Work with GEF (global environmental facility) Partner Agency to submit project proposals. Projects must be driven by the country, consistent with national | contact grant (request) | none | | | Special Climate Change Fund (SCCF) | The Special Climate Change multilateral climate adaptati created at the 2001 Confe |
| Green Climate Fund | World Bank | To receive money directly from the Green Climate Fund, entities need to be accredited as an "Accredited Entity" or AE. AEs can | Accredited Entities are responsible for monitoring and evaluating the projects they implement, as well as reporting their activities to GCF | contact grant (request) | none | | | Green Climate Fund (GCF) \| NDC Partnership | Set up in 2010 and aims to towards low-emission and c pathways by providing supp limit or reduce their greenhou to the impacts of |
| Green Climate Fund - Project Preparation Facility | World Bank | | Funding proposals developed with the support of the Project Preparation Facility should be submitted to the GCF Board within two years of the approval of a PPF request. | to USD 1.5 millio | none | | | Green Climate Fund (GCF) - Project Preparation Facility | Set up in 2010 and aims to towards low-emission and c pathways by providing supp limit or reduce their greenhou to the impacts of |
| Water Resilience Trust Fund under the Water Financing | Asian Development Bank (ADB) | ADB website for the fund: https://www.adb.org/what-we-do/funds/wfpf- | Support: Direct Project (preparation and implementation) Program quality (policies, | contact grant (request) | contact grant | | | WFPF-WRTF | The fund aims to support c actions by government age private sector needed to stren and the |

# Contributions

## White Paper

Enhanced the presentation of research findings, making them more accessible and engaging for stakeholders.



# Contributions

## Outreach

Engaged with organizations like the Vietnam USAID. Expanded professional relationships, strengthened the organization's network with key agency.

**Yen Nguyen** <yen.nguyen@flowmetric.io>
to Uyen

Dear Mrs. Uyen,

Thank you for your prompt response and for accommodating a quick phone call regarding the project information. We appreciate your time, especially given your busy training schedule.

Please let me know your availability and preferred time slots.

Thank you again for your assistance!

Best regards,

flowmetric

*Yen Nguyen*

yen.nguyen@flowmetric.io

https://www.flowmetric.io/

**Yen Nguyen** <yen.nguyen@flowmetric.io>
to Mbirnbaum, wilder.mccoy, nam.doan, Uyen

Dear Mr. Birnbaum,

I hope this message finds you well.

I would like to extend my sincere thanks to Mrs. Uyen for facilitating this connection.

Uyen mentioned that you will be in HCMC at the end of July. We would greatly appreciate the opportunity to meet with you in person. Please let us know your availability for a meeting.

We look forward to your response.

Best regards,

flowmetric

*Yen Nguyen*

yen.nguyen@flowmetric.io

https://www.flowmetric.io/

# Achievements & Lessons

I have gained a deep understanding of the challenges that new businesses face, with those unique models like ours focused on water security and sustainability.

- Able to conduct comprehensive research and analysis.
- Deepened understanding of the climate risk management space, especially in Southeast Asia.
- Worked closely with team members on various projects, enhancing teamwork and communications skills.

# Challenges

One of the most significant challenges I faced was adapting to a new industry and understanding its complexities. I initially felt overwhelmed by the technical terminology and intricate market dynamics. But investing time in reading industry reports and seeking guidance from my supervisor helps me learn and understand our company's structure.

| Risk Typology |
|---|
| *Water Quality Risk* |
| *Water Scarcity Risk* |
| *Flood Risk* |
| |
| *Water Inefficiency* |
| *Compliance Risk* |
| *Repuational Risk (Loss)* |
| *Repuational Risk (Gain)* |

| Water Tariff | M/3 of water Consumed |
|---|---|
| $ 0.51 | 146,350 |
| $ 0.76 | 146,350 |
| $ 1.01 | 146,350 |
| $ 2.01 | 146,350 |

| Water Tariff | M/3 of water Consumed |
|---|---|
| $ 0.51 | 146,350 |
| $ 0.76 | 146,350 |
| $ 1.01 | 146,350 |
| $ 2.01 | 146,350 |

# Recommendations

- Start projects early and take the time to explore host company and the related industry.
- More time you invest in understanding the company's operations and the industry's nuances, the more rewarding and impactful the experience will be.
- Opportunities for cross-departmental collaboration could provide a broader perspective.

# Reflection

After completing this internship, I plan to:

- Continue school
- Learning more about water finance, projects, and coursework

I am eager to continue learning about these topics, whether through coursework related projects at my university or future involvement with Flowmetric initiatives.